Predicting Default with Machine Learning models
PART 1: Technical Report

Marjorie Hirschi

Faculty of Computing & Data Sciences, Boston University

DX799 O2: Data Science Capstone

Joshua Von Korff, Ph.D.

April 19th, 2026

Executive Summary

The aim of this project is to 1. predict loan default risk and 2. identify the key factors and predictors of borrowers likely to default using Machine Learning modelling. This is critical, as the pillar of a profitable financial institution is its success at predicting default risk effectively, often referred to as Probability of Default. To achieve this, the project utilizes a large subset of data from the peer-to-peer lending platform Lending Club, which is available on Kaggle (George, 2018). Using approximately 400,000 loan records and 150 loan-related features, I developed several predictive models and refined the best-performing ones. The strongest-performing models, Elastic Net Logistic Regression and Gradient Boosting, correctly classified approximately 91–92% of borrowers who ultimately defaulted. Because defaults are relatively rare events, my modeling strategy prioritized Recall over Accuracy to ensure that high-risk borrowers were detected effectively.

In addition to supervised prediction, I tested several clustering algorithms to uncover possible hidden structural borrower segments in the data. K-Means was shown to produce the best results. K-Means uncovered four major risk clusters defined by home ownership status, income level, loan size, and loan term. Borrowers with mortgages tend to have higher incomes and lower default rates, while renters generally are generally less ‘credit worthy’.

In summary, I developed two satisfactory supervised models for predicting default (test recall: 92%, meaning that 92% of defaulting borrowers are correctly identified as defaulting), as well as an unsupervised model that identified structural aspects of borrower profiles that are directly transferable to other credit-related datasets. These models, taken together, produce a holistic framework for credit risk prediction that can be adapted or extended to other credit datasets.

Introduction and Problem Definition

Predicting borrower default is the cornerstone of credit risk modeling. Financial institutions derive their profit margins through an accurate evaluation of risk, which allows them to avoid financial losses, allocate capital efficiently, and comply with regulatory standards. Misclassifying high-risk borrowers as safe can result in substantial credit losses, while lending practices that are too conservative might erode profitability and limit access to credit. Gaining a better understanding of what causes and how to anticipate defaults would benefit the entire banking system and the broader economy. Peer-to-peer lending platforms such as Lending Club generate real-world, high-dimensional credit datasets that offer an opportunity to apply machine learning methods to default prediction.

Problem Definition

The core focus of this project is to build predictive models that help financial institutions speed up the loan approval process by accurately forecasting whether a loan will be Charged Off or Fully Paid at any stage of the loan's lifetime. Accordingly, this project pursues two related goals:

1. Identify the most influential factors associated with borrower default.
2. Develop a model that is adaptable and scalable to other credit datasets.

Research Questions

Thus, my research questions are directly derived from the problem definition:

- Which borrower and loan characteristics most strongly predict default?

- How much noise is embedded in loan datasets (i.e., which features are irrelevant and should be disregarded to help streamline the loan approval process)?
- Do unsupervised clustering methods reveal meaningful borrower segments associated with varying levels of default risk?
- Which models best predict default given the tabular nature of credit datasets? Will regularized linear models perform better, or will nonlinear ensemble methods outperform them in predicting default?

Success Criteria

Defaults represent the minority/positive class. The main focus of this project is to accurately predict defaulters, as missing them can be costly and may jeopardize the profitability and, thus stability of a financial institution. Consequently, I focused on maximizing the Recall and ROC-AUC metrics to evaluate model performance. The Recall rate quantifies the proportion of actual positive cases (defaulters) correctly identified by the model, while the ROC-AUC score evaluates the model's capability to discriminate between positive and negative classes (Singh, 2024). After refining the ML models I developed to analyse this dataset, I achieved a Recall of 0.90 or more for detecting defaulters, and maintained elevated discrimination with a ROC-AUC rate of 0.90 or more.

Scope and Constraints

Given the large size and depth of features of the Lending Club dataset, I focused exclusively on a subset sample (circa 400,000 rows, 152 features) and stopped looking at the other credit datasets that were allocated to me in the previous semesters. I also purposely narrowed my focus to

information available at loan origination and disregarded post-origination features, such as total_principal_received or total_late_fee_received in order to avoid data leakage.

Data Overview

The dataset consists of historical loan records from Lending Club, a peer-to-peer lending platform. The data includes borrower demographics, credit history variables, loan characteristics, and loan outcomes. As explained above, the dataset contains a pre-filtered subset of approximately 400,000 loans with 152 features. The data reflects real-world lending outcomes.

Table 1

Data Description

Observations:	400,000 loans
Features:	152 variables
Target variable:	loan_status (default or not)
% Null values:	31.35%
Data types:	**Typical tabular data** Numerical (loan amount, interest rate, income, FICO scores) Categorical (grade, sub-grade, home ownership, purpose)

Table 2

Data Dictionary Extract – Selected Variables/Features

Variable	Type	Description
loan_amnt	***Numeric***	Amount of the loan issued
int_rate	***Numeric***	Interest rate assigned
annual_inc	***Numeric***	Borrower's reported annual income
grade	***Categorical***	Loan grade (A–G)
sub_grade	***Categorical***	More granular loan grade
home_ownership	***Categorical***	Rent, Mortgage, Own
emp-length	***Categorial/Object***	Borrower's employment length in years
dti	***Numeric***	A ratio calculated using the borrower's total monthlu debt payments on the total debt obligations, excl. mortgage, divided by the borrower's monthly income
num_op_rev_tl	***Numeric***	Number of open revolving accounts
term	***Numeric***	Loan duration (months)
loan_purpose	***Categorical***	Purpose of loan

From the summary statistics table below (Table 3), we can infer that the average borrower earns an annual income of $77k and takes out a loan of approximately $15k for a 42-month term at an average interest rate of 13.1%.

Table 3

Summary Statistics

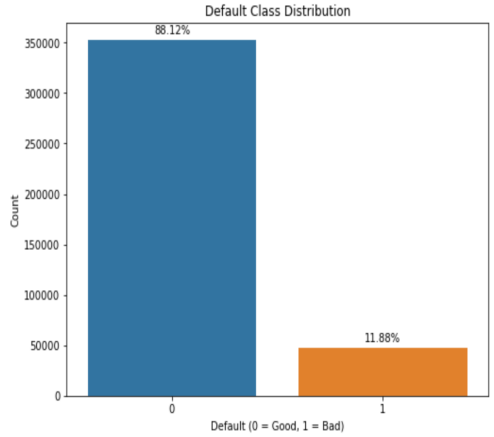
	member_id	loan_amnt	funded_amnt	funded_amnt_inv	term	int_rate	installment	annual_inc	dti
count	0.0	399996.000000	399996.000000	399996.000000	400000.000000	399996.000000	399996.000000	3.999950e+05	399695.000000
mean	NaN	15042.661614	15037.214622	15019.838961	42.920040	13.092479	445.557073	7.777201e+04	18.859016
std	NaN	9194.553141	9192.507485	9196.881776	10.872714	4.829744	267.143635	7.290914e+04	14.260549
min	NaN	600.000000	600.000000	0.000000	0.000000	5.310000	4.930000	0.000000e+00	0.000000
25%	NaN	8000.000000	8000.000000	8000.000000	36.000000	9.490000	251.580000	4.600000e+04	11.920000
50%	NaN	12887.500000	12850.000000	12800.000000	36.000000	12.620000	377.870000	6.500000e+04	17.860000
75%	NaN	20000.000000	20000.000000	20000.000000	60.000000	15.990000	593.060000	9.300000e+04	24.520000
max	NaN	40000.000000	40000.000000	40000.000000	60.000000	30.990000	1719.830000	9.500000e+06	999.000000

Exploratory Data Analysis

I performed a detailed EDA to investigate feature distributions, identify class imbalances, analyse relationships between predictor variables and the target, as well as detect outliers. During the EDA, I discovered the presence of potential data leakage: several features contained post-loan outcome information, which could unfairly influence model predictions. I dropped these features in the Data Cleaning phase to ensure that only data available at the time of loan origination was used. As the main objective of this project is to predict default, I converted the loan_status target feature, which contained multiple categories, into a binary variable: I encoded loans labeled as ‘Fully Paid’ and other non-default statuses as 0 (No Default), and those marked as ‘Charged Off’ as 1 (Default). Following this, it became clear that there was an important class imbalance in the target variable: the vast majority of loans were good/not charged off, in line with real-world conditions in which defaults are relatively rare (see Figure 1 below).

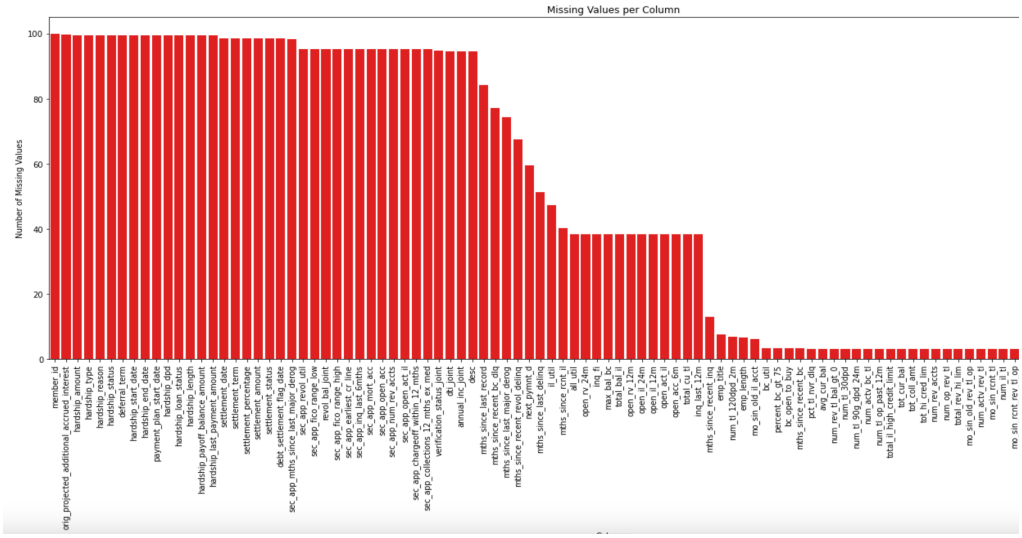
Figure 1

Default Class Distribution



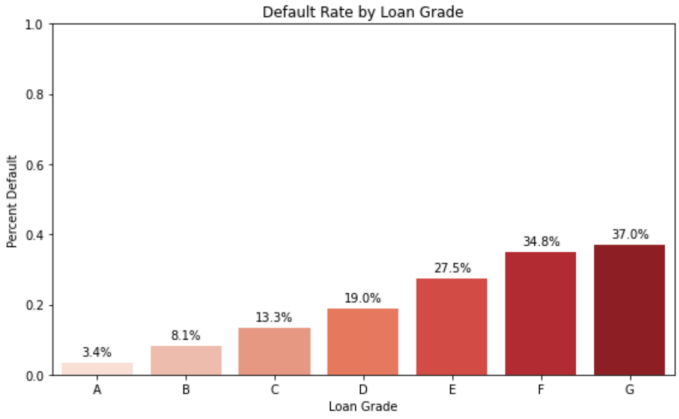
After investigating missing values, I discovered that the dataset contained numerous features with more than 65% missing values, as illustrated in Figure 2. This bar chart clearly shows that the percentage of missing values per feature drops sharply around the 65% threshold and even more noticeably after the 40% threshold. Thus, I removed features with 65% or more missing values. I then imputed the remaining missing values using the median, as many features exhibited skewed distributions making the median a more appropriate measure of central tendency than the mean.

Figure 2 - *Missing Values per Variable*



I addressed outliers by focusing primarily on extreme income values, which I considered strong indicators of potential data inconsistencies. I also removed highly multicollinear features, as they are inherently redundant. For example, variables such as funded_amnt, installment, and the sub-versions of the main FICO rating feature, more granular variants of the primary credit score, were excluded from the analysis. While higher loan amounts showed a modest positive correlation with default, loan grades (and sub-grades), as well as interest rates displayed a much stronger relationship and emerged as important predictors of default. Loan Grade and Sub-grade were particularly informative, with default rates increasing consistently with lower-grade loans (e.g., F and G). This pattern can be seen in the Figure 3 below titled Default Rate by Loan Grade.

Figure 3 - *Default Rate by Loan Grade*



Finally, I split the dataset into training and test sets, standardized the features to prevent variables with larger magnitudes from disproportionately influencing the models, and started building several machine learning models.

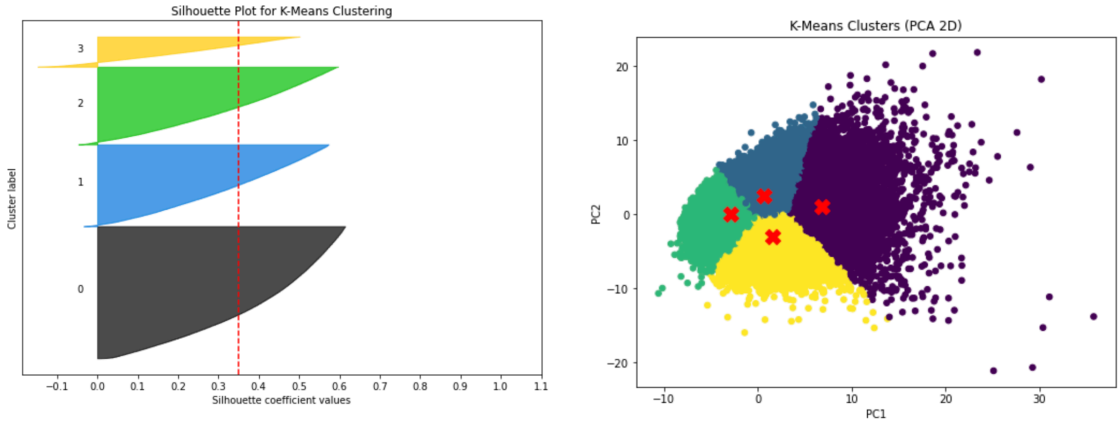
EDA using Unsupervised Learning

I used various clustering models as a data exploration tool (such as DBSCAN, HAC) to identify underlying patterns in the dataset. Of all these models, K-Means was the only model that generated relevant clusters. Since K-Means is a distance-based algorithm (using Euclidean distance), it is sensitive to the scale of the features, particularly in high-dimensional spaces. To address this, I first scaled the features using StandardScaler and then reduced the dataset's dimensionality to two principal components using PCA. Because the number of clusters (k) is a hyperparameter of K-Means, I selected the optimal value using:

- the Inertia metrics, that is, the mean squared distance between each instance and its closest centroid (with the Elbow Method) (Géron, 2019), and
- the Silhouette scores, which measures separation and cohesion of clusters (Géron, 2019).

I computed both metrics for k = 2 to 10. The elbow curve showed a gradual decrease but no sharp "angle," so no obvious cluster count seemed apparent. Silhouette scores peaked around k = 4 to k = 6. I used k=6 and k=4 to produce two silhouette graphs/diagrams. Based on the shapes of the clusters represented in the silhouette diagrams and the vertical dashed line indicating the silhouette score, I decided to focus on four clusters (Géron, 2019): the silhouette score was higher for them - see silhouette diagram below for four clusters.

Figure 4 - *K-Means Silhouette Plot and 2D Visualization using PCA*



The most paramount conclusion I derived from the four-cluster analysis (see Table 4 below) was that the likelihood of 'default' primarily depends on whether a borrower rents, has a mortgage, or owns a home.

Table 4 - *K-Means: Four-cluster analysis*

Cluster	Home Ownership Profile	Avg. Income	Avg. Loan Size	Loan Term	Default Rate	Segment Insight
Cluster 1	46% Mortgage - **42% Rent** - 12% Own	$69k	$11.3k	41 months	**18% (Highest)**	Mixed housing status and mid-income borrowers; smaller loans but **highest default risk**.
Cluster 2	**54% Rent** - 34% Mortgage - 12% Own	$56k	$9.9k	40 months	**12%**	Renter-dominant, lower-income group taking the smallest loans; moderate risk.
Cluster 0	**70% Mortgage** - 18% Rent - 12% Own	**$124k**	**$22.5k**	46 months	**11%**	High-income borrowers taking relatively large loans with longer terms; moderate risk.
Cluster 3	**69% Mortgage** - 22% Rent - 9% Own	**$101k**	**$23.5k**	**47 months**	**6% (Lowest)**	High-income mortgage holders with the largest loans and longest terms; **lowest risk segment**.

From the above, I concluded that clusters with high rates of mortgage ownership (0 and 3) tend to have higher incomes, larger loans, longer terms, and lower default rates. Conversely, rent-heavy clusters (2 and partly 1) generally have lower incomes, smaller loans, shorter terms, and higher default rates, especially cluster 1. In summary, mortgage ownership correlates with financial stability, renting relates to smaller loans and slightly increased risk, while longer loan terms are associated with more stable clusters. The below graphs (Figure 5 & 6) confirm

that clusters differ materially in their risk profiles, confirming that my final K-Means model with four clusters, uncovered meaningful, structural subgroups among borrowers.

Figure 5

Loan Amount Distribution Across Clusters

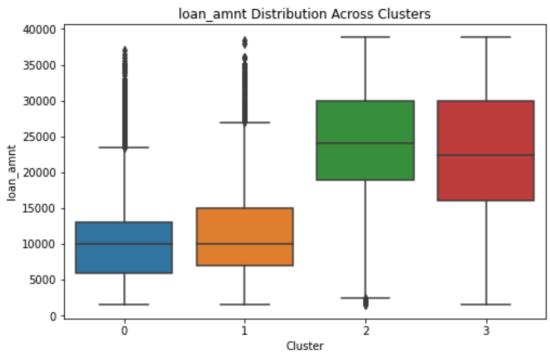
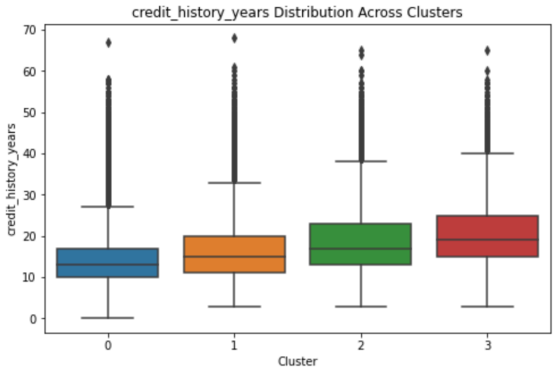


Figure 6

Credit History Distribution Across Clusters



From Correlation to Causal Inference: Propensity Score Matching

While the K-Means clustering analysis revealed that rent-heavy clusters exhibit higher default rates, these results remain correlational. Borrowers who rent differ from those with mortgages across several variables, including income, loan size, and credit history. To further investigate whether home ownership status itself is associated with default risk,independent of these confounding variables, I propose extending the analysis using Propensity Score Matching (PSM) going forward. PSM is a technique used to reduce selection bias in observational datasets by constructing a control group that is statistically similar to the treatment group (Huntington-Klein, 2021, Chapter 14.5). In this context:

- Treatment group: Borrowers who rent
- Control group: Borrowers with mortgages

In the context of this analysis, the propensity score would be the probability of being a renter conditional on observed characteristics (Huntington-Klein, 2021, Chapter 14.5). I would estimate this probability using a logistic regression model, where the dependent variable is home ownership status (i.e renter vs. mortgage holder), and the independent variables include key borrower and loan attributes such as Annual income, Loan amount, Loan term, Debt-to-income ratio and Credit history indicators (e.g., FICO score). Using these estimated propensity scores, renters would be matched to mortgage holders with similar profiles (e.g., via nearest-neighbor matching). This would create a balanced sample in which the two groups are comparable across covariates. Once matched, I would compare default rates between the two groups to estimate the Average Treatment Effect on the Treated (ATT) (Huntington-Klein, 2021). This would allow to answer this refined research question: Among borrowers with similar financial profiles, does renting remain associated with a higher probability of default? If the difference in default rates remains statistically significant after matching, this would suggest that renting is independently associated with higher credit risk. Conversely, if the difference decreases or disappears, it would indicate that the higher default rates observed in the clustering analysis are mainly driven by other underlying financial characteristics rather than home ownership status itself.

Ethical Considerations

As noted above, the K-means clustering analysis shows that rent-heavy clusters tend to pose a higher default risk. However, I must warn against drawing conclusions too quickly given the following two weak points:

1. As explained in the Correlational to Causal Inference section above, at this time of my analysis, I can only conclude that the observed relationship highlighted by the K-Means analysis is purely correlational, as a causal link has not yet been established.
2. The underlying data structure likely contains structural historical biases causing the correlations and clustering resulting from K-Means. While the Lending Club dataset does not contain personally identifiable information (variables such as race or gender are prohibited by law in credit datasets), potential biases may still be reflected due to the historical data such datasets contain. For example, features such as FICO scores or debt-to-income ratios can serve as indicators of wealth accumulation and depth of credit history. These indicators tend to correlate with socioeconomic status, which in turn correlates with ethnicity, race, or gender. In other words, certain groups of borrowers, in particular, minorities and women, often have "thinner" credit histories. This occurs because some sub-groups rely primarily on cash transactions or a limited number of credit cards (one or two), which inadvertently hinders the development of a solid credit profile. In addition, structural inequalities in access to wealth-building opportunities can compromise wealth accumulation even for high-income individuals. This limited data can lead credit models to wrongfully label such borrowers as "bad credit", despite their potential as reliable, high-quality borrowers.

Thus, to remediate these biases, I am proposing adding high-quality alternative data to this dataset, such as rent payment history by, for example, requesting recommendation letters from landlords or other alternative sources that would show good standing on rent payment and help create more granular clusters in the K-means analysis performed above. Additionally, applying Propensity Score Matching as I proposed above, could potentially help mitigate bias by

distinguishing between true risk factors and proxies for socioeconomic status. If renting is no longer associated with higher default risk after matching, this would suggest that models relying heavily on home ownership may inadvertently penalize borrowers due to structural inequalities rather than actual creditworthiness.

Finally, features that introduce data leakage, such as inadvertently including post-default or post-approval information, can also create bias which is another reason why I made the decision to remove them.

Modeling and Analysis

While I experimented with many models, including Ordinary Least Squares, K-Nearest Neighbors, SVM, and Random Forest, in this section, I focused only on the experiments and results that led to my two best-performing models: Elastic Net Logistic Regression and Gradient Boosting Classifier.

Regularized Logistic Regression with Lasso, Ridge, and Elastic Net Penalties

I started by exploring some of the simplest and most interpretable models available in the supervised Machine Learning toolbox: Linear Models. Thus, I started my modeling journey by implementing Logistic Regression with Lasso, Ridge, and ElasticNet regularization. I evaluated these models using the typical classification metrics (i.e Recall, F1, Accuracy ROC AUC), and put a specific emphasis on maximizing Recall and ROC AUC (as explained above). I also used the hyperparameter, ‘class_weight:balanced’, to handle the imbalanced target feature described above (GeeksforGeeks, 2024).

Model Optimization and Evaluation

I performed GridSearchCV to identify the optimal hyperparameters for each model, using Logistic Regression as the base estimator. I evaluated all three regularization types (L1, L2, and Elastic Net) and tuned the regularization strength parameter C. For Elastic Net, I additionally tuned the l1_ratio parameter. Model selection was based on the highest recall score across cross-validation folds. As already explained, I prioritized Recall because detecting true defaulters was more important than minimizing false positives. Results are shown in the Table 5 below:

Table 5

Metric	Logit	Lasso	Ridge	ElasticNet
CV Accuracy	0.915 ± 0.001	0.915 ± 0.001	0.915 ± 0.001	0.915 ± 0.002
CV Precision	0.526 ± 0.004	0.527 ± 0.003	0.527 ± 0.004	0.528 ± 0.005
CV Recall	0.914 ± 0.011	0.914 ± 0.011	0.914 ± 0.011	0.915 ± 0.011
CV F1 Score	0.668 ± 0.005	0.669 ± 0.004	0.668 ± 0.005	0.669 ± 0.005
CV ROC AUC	0.964 ± 0.003	0.964 ± 0.003	0.964 ± 0.003	0.964 ± 0.003
Test Accuracy	0.9133	0.9135	0.9133	0.914
Test Precision	0.5387	0.5396	0.539	0.5409
Test Recall	0.9127	0.912	0.9124	0.9154
Test F1 Score	0.6776	0.678	0.6777	0.68
Test ROC AUC	0.964	0.9641	0.964	0.964

Elastic Net was the top performer of all the models evaluated during GridSearchCV, with Test Recall at 91.5%, Test F1 Score at 68.0%, Test ROC AUC at 0.964 and Test Accuracy at 91.4%.

Feature Engineering with Polynomial & Interaction Terms

Once I had reached the best 'optimized' Logistic Regression model, I investigated adding both a polynomial ('int_rate_squared') and an interaction term ('loan_amount' * 'annual inc'). I then retrained this 'best' model (i.e, with Elastic Net penalty, C=10, l1_ratio=0.5) and found out that output results stayed in line with what I had before, but at slightly lower levels: test Recall stood at 91.2%, test F1 at 67.7%, ROC AUC at 0.964 with precision at 53.8%. Unfortunately, these

newly engineered features did not add any predictive value, indicating that the regularized logistic model already captured effectively most of the predictive information within the data.

Feature Selection

I implemented Forward Feature Selection using repeated 5-fold cross-validation. Given the large size of the dataset, it was computationally expensive and took a long time to complete. Consequently, I did not conduct Backward Feature Selection. However, after running the Forward Feature Selection algorithm, the best features (65 in total) were output. I re-ran my best Logistic Regression model with the selected features only and found very similar results to those shown above. In other words, Forward Feature Selection did not add predictive value, as the ElasticNet regularization term had already "inherently" performed feature selection. Finally, I applied PCA to reduce the dimensionality of the feature space, retaining 99 principal components that explained 95% of the variance. Logistic regression was then trained on the transformed data. Surprisingly, while the model achieved high overall accuracy (94%) and strong performance on the majority class, the recall for the minority class (defaulters) dropped to 63%, compared to a 91% recall on the test set for the Elastic Net Logistic Regression model described above. Thus, PCA combined with Logistic Regression did not appear to add predictive value, as Logistic Regression with the Elastic Net penalty already performed effective feature selection by shrinking unimportant weights/coefficients. When combined with PCA, it seemed that the model underfitted the data, resulting in poorer performance.

Gradient Boosting Classifier

As for Logistic Regression, I tuned Gradient Boost using 5-fold stratified CV, thus reducing overfitting risk by averaging performance across folds. Again, I optimized for Recall since capturing defaults is the primary business goal. I also performed a randomized parameter search (30 iterations) exploring learning rate, depth, leaf constraints, and minimum leaf sizes. However, I did not initially re-balance the dataset, as it appeared that Gradient Boosting (as per Sklearn) did not include a dedicated class-weighting parameter (unlike models such as logistic regression or random forests). Consequently, the classification report showed high overall accuracy but poor recall for the minority class (circa 0.70). After investigating how to adjust class weights manually (GeeksforGeeks, 2025), I chose to apply the standard balanced weighting formula provided in the Sklearn documentation (Scikit-learn, 2025):

$$W_i = n_{samples} / (n_{classes} \times n_i),$$

where n_samples is the total number of samples, n_i is the number of samples in class i,and n_classes is the number of classes.

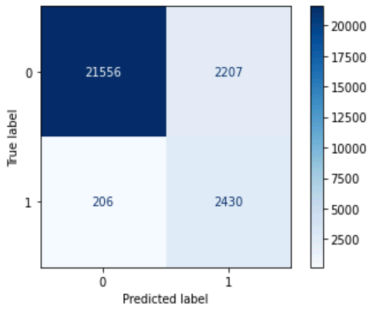
According to Sklearn, the correct way to handle class imbalance for Gradient Boosting is to pass the sample weights calculated with the above formula directly into the fit() method. After doing so, I finally improved the model's recall to 0.92, matching the performance of the logistic regression (with Elastic Net regularization). ROC-AUC also improved to approx. 0.94 (see Classification Report below).

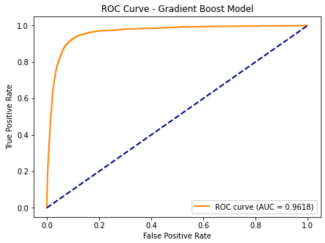
Figure 7 to Figure 9

Classification Report (Test Set), Confusion Matrix & ROC Curve for Gradient Boosting

Classification report on test set:

	precision	recall	f1-score	support
0	0.99	0.91	0.95	23763
1	0.52	0.92	0.67	2636
accuracy			0.91	26399
macro avg	0.76	0.91	0.81	26399
weighted avg	0.94	0.91	0.92	26399





What surprised me was that Gradient Boosting did not outperform the Logistic Regression model. Instead, it achieved comparable performance. This was somewhat unexpected, as Gradient Boosting is generally better at capturing complex nonlinear relationships, although at the cost of reduced interpretability. One possible explanation is that the data's underlying structure is largely linear, which implies that a simpler model was sufficient to capture the signal.

Results Summary

Model	Recall	ROC-AUC	Accuracy
Elastic Net Logistic	91.5%	0.964	91.4%
Gradient Boosting (weighted)	92%	~0.94	91%

Conclusion/Recommendations

Both Gradient Boosting and Elastic Net Logistic Regression were my top-performing models, with a test recall of approx. 91.5% - 92%, and a ROC AUC score of 0.964 and 0.94, respectively. Despite the slightly higher ROC AUC of Gradient Boosting, my preference goes to the Elastic Net Logistic Regression model as it balances prediction and interpretability. The Elastic Net regularization (a combination of L1 and L2 penalties) improved the model by shrinking uninformative features, enabling 'inherent' feature selection, thus preventing overfitting. In its

current form, the model delivers strong predictive performance and effectively addresses the primary business objective: identifying potential defaulters. With over 91% recall and comparable accuracy, it demonstrates reliable classification capability.

In parallel, my K-Means Clustering analysis revealed tangible borrower segments, particularly highlighting the role of home ownership and income in risk differentiation. Thus, K-Means helped address the second objective of this project: identifying the most influential factors associated with borrower default. To further refine these clusters and promote lending fairness, I recommend incorporating non-traditional credit data, such as verified rent payment history into future versions of the dataset. This would ensure a more granular assessment of reliable borrowers located in the 'rent-heavy" segments, who may currently lack a strong credit profile but could appear in their own cluster of "low default" renting borrowers. Additionally, I propose conducting Propensity Score Matching going forward, as it could potentially help mitigate bias by distinguishing between true risk factors from proxies for socioeconomic status. For example, if renting is no longer associated with higher default risk after matching, this would suggest that current models are capturing structural inequalities rather than true creditworthiness.

Beyond refining model parameters, I recommend revisiting EDA and exploring clustering-informed modeling approaches, where separate models are trained on borrower clusters/segments and their predictions are combined (e.g., via Ensembling). This approach could improve recall and precision by capturing segment-specific patterns that a single global model may overlook. As a further extension, a neural network could be explored to capture complex non-linear relationships and higher-order feature interactions.

In summary, I believe that the modeling framework proposed in this project is comprehensive, interpretable, adaptable, and well-aligned with the original research objectives, offering a robust foundation for scalable credit risk prediction.

References

George, N. (2018). *All Lending Club loan data: 2007 through current Lending Club accepted and rejected loan dataset*. Kaggle. https://www.kaggle.com/datasets/wordsforthewise/lending-club

Singh, A. (2024, October 3). *ROC Curve In Machine Learning?* https://www.appliedaicourse.com/blog/roc-curve-in-machine-learning

Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (2nd ed.). O'Reilly Media. Chapter 9: Unsupervised Learning Techniques – K-Means (silhouette score and diagram).

GeeksforGeeks. (2024, March 11). *Weighted logistic regression for imbalanced dataset*. https://www.geeksforgeeks.org/machine-learning/weighted-logistic-regression-for-imbalanced-dataset/

Brownlee, J. (2020, October 30). *Cost-sensitive logistic regression for imbalanced classification*. MachineLearningMastery.com. https://machinelearningmastery.com/cost-sensitive-logistic-regression/

GeeksforGeeks. (2025a, July 23). *Bagging and Random Forest for imbalanced classification*. https://www.geeksforgeeks.org/bagging-and-random-forest-for-imbalanced-classification/

Huntington-Klein, N. (2021). *The Effect: An Introduction to Research Design and Causality* (1st ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9781003226055